

Applying Variational Quantum Classifier on Acceptability Judgements: A QNLP Experiment

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Abstract. The newborn Quantum Natural Language Processing (QNLP) field has experienced tremendous growth in recent years. The possibility of applying quantum mechanics to critical aspects of language processing has dramatically impacted many tasks, ranging from theoretical approaches to algorithms implemented on real quantum hardware. From a methodological point of view, the possibility offered by applying quantum mechanics to NLP problems is well suited to classification tasks. This work aims to test the potential computational advantages of a hybrid algorithm, namely the Variational Quantum Classifier (VQC), to perform classification on a classical Linguistics task: acceptability judgments. VQC is a quantum machine learning algorithm able to infer the relations between input features and the associated belonging class using a parametrized quantum circuit and an encoding layer that embeds classical data into quantum states. An acceptability judgment is defined as the ability to determine whether a sentence is considered as natural and well-formed by a native speaker. The approach has been tested on sentences extracted from ItaCoLa, a corpus that collects Italian sentences labeled with their acceptability judgment. The evaluation phase has considered quantitative metrics and qualitative analysis to investigate further the algorithm's behavior on specific linguistic phenomena included in the corpus.

Keywords: Quantum Machine Learning · Quantum Natural Language Processing · Variational Quantum Classifier

1 Introduction

Recently, there were significant advancements in all Natural Language Processing (NLP) tasks, ranging from machine translation [41], text classification [34], coreference resolution [10,12,19,28] or multi-language syntactic analysis [5,13,14]. This is primarily due to the explosion of neural language models based on deep learning architecture, especially Transformers-based models such as BERT [7]. However, this improvement comes with an increasing complexity of models, which requires a considerable amount of data and computation resources

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Y. D. Sergeyev et al. (Eds.): NUMTA 2023, LNCS 14478, pp. 98–112, 2025.
https://doi.org/10.1007/978-3-031-81247-7_8

to be efficiently trained [3,9]. Besides, there are open issues related to what these models learn about language, how they encode this information [15], and how much of it is interpretable [17]. Quantum machine learning (QML) is gaining attention as an alternative approach that exploits powerful aspects borrowed from quantum mechanics to overcome the computational limitations of current approaches. A strand derived from QML is the sub-field of Quantum Natural Language Processing (QNLP) [6], which aims to solve natural language-related tasks using quantum properties or applying algorithms derived from quantum theory or testing new approaches using real quantum hardware.

However, quantum-based approaches currently suffer the limitations of available hardware, and there are open issues about scalability because the quantum circuit can be considered a linear or at most sub-linear model in the feature space [31]. There is an ongoing research effort concerning the encoding of classical data into quantum computers, which in principle allows to extract information and make computations more efficiently than classical computer, but with a strong backside when considering large dataset or data represented as large vectors in feature space. Nowadays hardware has a limited amount of qubits and, when the number of qubits is sufficient for embedding data, the lack of fault tolerance, i.e., the quantum noise associated with the computation, hinders the performances and destroys the potential advantages of quantum computing. To overcome these limitations, novel hybrid approaches combining classic pre-trained models and quantum techniques have been proposed [20,21]. This type of approach offers the advantage of implementing specific layers of models on a quantum device, while classical models for non-linear operations handle intermediate results. A successful example of such hybrid approaches is the so-called classical-quantum transfer learning [24]. It encodes input features in a multi-qubit state, then a quantum circuit transforms and measures such features. In this pipeline, output probabilities are projected to the task label space, and losses are backpropagated to update parameters.

Starting from the approach proposed in [21], this work proposes a hybrid transfer learning model for QNLP applied to a binary classification task. It uses the pre-trained Sentence-BERT model and fine-tunes it to perform the classification. This approach has been tested on current noisy intermediate-scale quantum (NISQ) machines [35]. It exploits the advantage and robustness of classic pre-trained language features already well-known in the literature [29] and integrates quantum encodings to reduce noisy devices. As already demonstrated in [21], avoiding tensor product of vectors is useful to allow a scalable QNLP model and improve representativeness facing natural language sentences of various length and complexity. The Variational Quantum Classifier (VQC) [4] is the algorithm chosen to perform a binary classification on acceptability judgments, a task which has gained much popularity in recent years in the field of NLP [22,39]. This particular task has been chosen because QNLP has proven particularly effective in binary classification [26,33]. For the purpose of this work, which focuses on the Italian language, the dataset chosen has been ItaCoLa [2,36]. It is the largest existing resource for this task in Italian, collecting sentences labeled

with their judgments by expert linguists. The evaluation phase has considered quantitative metrics.

The paper is organized as follows: in Sect. 2, the research works available in the literature related to what is presented are described, while in Sect. 3 the dataset and applied methodologies are presented, then in Sect. 4, the results obtained are exposed, and relevant aspects are discussed, and finally, overall conclusions are drawn in Sect. 5.

2 Related Work

2.1 Quantum Machine Learning

Numerous examples have received increasing interest in recent years concerning the adaptation of classical machine learning algorithms through the use of properties and techniques borrowed from quantum mechanics.

One notable approach is Quantum Support Vector Machines (QSVMs), which aim to enhance the performance of traditional Support Vector Machines (SVMs) by utilizing quantum algorithms. QSVMs have shown promising results in various text classification tasks, including sentiment analysis, topic classification, and document classification. Another area of research is quantum-inspired algorithms for text classification. Quantum-inspired algorithms, such as Quantum-Inspired Genetic Algorithm (QGA) and Quantum-Inspired Particle Swarm Optimization (QPSO), draw inspiration from quantum mechanics and apply quantum-like principles to improve traditional optimization techniques. These algorithms have been explored in the context of feature selection and parameter optimization for text classification, demonstrating their potential to enhance classification accuracy and efficiency. Moreover, quantum embeddings have gained attention as a means to represent and analyze textual data. Quantum embeddings leverage the concepts of quantum superposition and entanglement to capture semantic relationships between words or documents. These embeddings aim to capture more nuanced and context-dependent information compared to traditional word embeddings. By utilizing quantum representations, classification models can benefit from enhanced semantic understanding, improving performance in various NLP tasks.

Furthermore, quantum machine learning algorithms, such as Quantum Neural Networks (QNNs) and Quantum Boltzmann Machines (QBMs), have been explored in text classification. These quantum-inspired models leverage the unique computational capabilities of quantum systems to perform complex computations efficiently. Although still in its early stages, quantum machine learning holds the potential to address the computational challenges associated with large-scale language data and to provide more powerful models for classification tasks.

2.2 Quantum Natural Language Processing

The growing interest in exploring the potential of QML in language-related tasks has led to the birth of the QNLP. As a sub-field of QML, QNLP offers a new

paradigm focused on processing and analyzing language data by leveraging the principles of quantum mechanics.

QNLP approaches proposed so far can be tested on real datasets using classical hardware (quantum-inspired approaches) or on currently available quantum machines (quantum-computer approaches). Quantum-inspired approaches integrate the advancements of quantum mechanics into an existing model. They are usually tested on benchmark datasets surpassing the current state-of-the-art. By contrast, Quantum-computer approaches are tested on NISQ hardware. Due to the limitations of current quantum hardware, these approaches are tested on simple NLP tasks and only on small-to-medium scale datasets. Several studies have investigated the application of quantum computing in classification tasks within NLP; a detailed review is described in [11].

An alternative way proposed to push through the limitations inherent in the scalability of such experiments is represented by hybrid approaches. A hybrid classical-quantum scheme using a quantum self-attention neural network (QSANN) has been recently proposed [20]. Even if it introduces the possibility of non-linearity, significantly improving over other QNLP models [23], this approach is limited by the continuous switching between quantum and classical hardware at each self-attention layer needed to run the network. [21] proposes a more viable approach to solve the low non-linearity issue for QNLP models using the classical-quantum transfer learning paradigm [24]. Using the classical-quantum transfer mechanism and pre-trained quantum encodings seems the most promising approach to develop scalable QNLP models, paving the way for the possibility of being implemented on real quantum hardware. Given this high potential and the increasing appeal of this approach, it was chosen for use in this work.

2.3 Acceptability Judgements Task

Automatically assessing acceptability tasks has always been a popular task in Linguistics. However, first resources built have been limited to small dataset theoretical purposes or within psycholinguistic experiments [18,25]. Since the release of the CoLa corpus [39], the first large-scale corpus of English acceptability, containing more than 10k sentences, the task has garnered increasing interest from researchers in the field of NLP.

The CoLA corpus has been presented with a number of experiments to assess the performance of neural networks on a novel binary acceptability task. Furthermore, it has been included in the GLUE dataset [38], a very popular multi-task benchmark for English natural language understanding, and an acceptability challenge has been launched on Kaggle,[1]. For such reasons, the number of studies dealing with binary acceptability has remarkably increased, although proposed approaches have often used different metrics for evaluation, so comparisons cannot always be made.

Starting with the same methodology introduced in COLA, similar resources have been released in recent years in various typologically different languages,

[1] https://www.kaggle.com/c/cola-in-domain-open-evaluation/.

ranging from Italian [36] - which is the subject of this work - Norwegian [16], Swedish [37] and Russian [27], Japanese [32], and Chinese [40].

3 Materials and Methods

3.1 ItaCola Dataset

The dataset used for this work is ItaCoLA, the Italian Corpus of Linguistic Acceptability [36]. This corpus was designed to represent a broad range of linguistic phenomena while distinguishing between sentences that are considered acceptable and those that are not. The process used to create the corpus was modeled as closely as possible on the methodology employed for the English CoLA dataset [39]. ItaCoLA consists of approximately 9,700 sentences drawn from various sources that cover numerous linguistic phenomena.

Table 1. Example sentences from the ItaCoLA dataset. 1 = acceptable, 0 = not acceptable. The symbol $*$ is conventionally used in Linguistics to mark unacceptable sentences.

Label	Sentence
0	*Edoardo è tornato nella sua l'anno scorso città. (*Edoardo returned to his last year city)
1	Ho voglia di salutare Maria (I want to greet Maria)

The acceptability annotation for these sentences is based on Boolean judgments formulated by experts who authored the different data sources. The sentences are sourced from various linguistic publications spanning four decades, transcribed manually and released in digital format. An example of how the data are structured in the corpus is shown in Table 1. ItaCola is divided into training, validation, and test split, including respectively 7,801, 946, and 975 examples. In [36] a baseline using LSTM with FastText embeddings and an Italian version of BERT has been released.

3.2 Methodology

This section outlines the methodology for incorporating Italian BERT embeddings in Quantum Natural Language Processing (QNLP) pipeline for acceptability judgements classification task. The proposed pipeline is articulated in following steps:

– **Data Preprocessing**: Sentences extracted from ItaCola needs to be preprocessed and prepared for embedding generation.

- **Embedding Generation**: For the purpose of this work, pretrained Sentence-BERT [30] embeddings have been used. The embedding model is trained on a large corpus of text data to learn the semantic relationships between words, using siamese and triplet network to represent a semantically meaningful embedding for the a whole sentence. Specifically, Sentence-BERT is trained for the classification of pairs of sentences, where the scope of the learning procedure is to optimize a similarity score between sentences. This training process involves learning the contextual information surrounding each word in the corpus. Feature vectors extracted via Sentence-BERT have 768 real-valued entries. These values need to be encoded in a quantum state for training a quantum classifier.

- **Quantum Embedding Construction**: Quantum embeddings aim to leverage the principles of quantum mechanics to represent language data in a quantum state. These embeddings capture more nuanced and context-dependent information compared to classical embeddings. One approach to constructing quantum embeddings is to utilize quantum superposition and entanglement. The embeddings are generated by encoding the classical embeddings into quantum states, where the amplitudes represent the weights or probabilities associated with each word or document. Quantum circuits or operators are employed to perform transformations on the quantum state, enabling the manipulation and analysis of the quantum embeddings.

- **Quantum Classification Model**: Once the quantum embeddings are constructed, they can be used as inputs to quantum classification models for various NLP tasks. In this work, the variational quantum classifier is constructed with Quantum-inspired neural networks (QNNs). These are generally composed of a series of single and two-qubit gates with free parameters that are trained in the learning phase. The structure of these circuits needs to be sufficiently complex to accommodate the possible solutions of the task, and yet simple enough to prevent detrimental effects from quantum noise. There is currently an ongoing research effort aiming to find the optimal strategy for the construction of a variational quantum circuits [8]: one of the key element to take into account is a certain level of entanglament, i.e. quantum correlation, between qubits so that information is shared among each element of the computation and complex solution are potentially explored during training. Here, a basic entanglement ansatz is used, where each qubit is going to be forced in a quantum correlated state with another qubit, pairwisely.

- **Model Training and Evaluation**: The quantum classification model, along with the quantum embeddings, is trained using a labeled dataset. This dataset consists of instances with their corresponding class labels. Model training involves optimizing the parameters of the quantum classification model to minimize a predefined loss function, such as hinge loss for SVMs or cross-entropy loss for neural networks. The trained model is evaluated using appropriate evaluation metrics, such as accuracy, precision, recall, and F1 score, to

assess its performance on unseen data.

– **Evaluation**: To assess the effectiveness of quantum embeddings in QNLP tasks, it is beneficial to compare their performance against classical embedding-based models. Classical models, such as traditional machine learning algorithms or neural networks, can be trained and evaluated using the same dataset and evaluation metrics. The performance of the quantum embedding-based model can be compared to these classical models to determine the impact and benefits of quantum embeddings in QNLP tasks. It is worth noting that the specific implementation details of the methodology may vary depending on the chosen embedding technique, quantum embedding construction approach, and the specific quantum classification model employed. The above steps provide a general framework for integrating embeddings in Quantum Natural Language Processing.

In this work three classification pipelines are used, so that the performances out of each strategy can be compared. A schematic of each model can be seen in Fig. 1 and a detailed explanation is provided in followings.

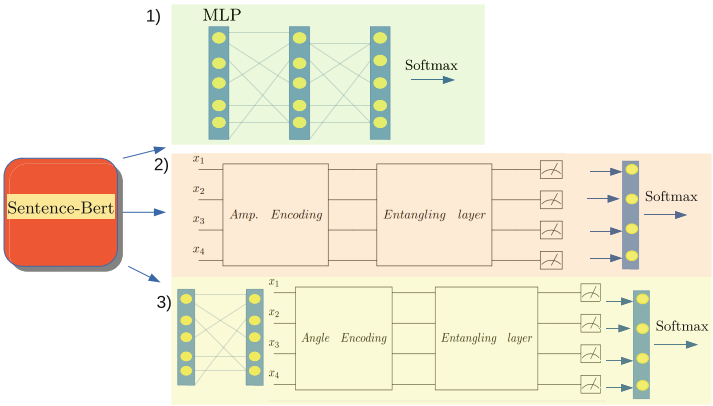


Fig. 1. Schematic of the different training strategies. In **1)** embeddings from Sentence-BERT are the inputs of a Multi-Layer perceptron, i.e. in a complete classical approach, trained for binary classification (**SentenceC**). In **2)** emebeddings vectors are encoded in the amplitude of a quantum state, then the information is passed to a parametrized quantum circuit. The outcome of the measurements is then used as input of a small Multi-Layer perceptron for binary classification (**SentenceQF**). In **3)** emebeddings vectors are first passed through a small Multi-Layer perceptron for dimensionality reduction and then encoded in the rotation angle of single-qubit quantum gates. Then the information is passed to a parametrized quantum circuit. The outcome of the measurements is finally used as input of a small Multi-Layer perceptron for binary classification (**SentenceQS**).

SentenceC. First a fully classical pipeline is implemented (which is in the following is named for convenience **SentenceC**). Here, the preprocessed dataset is converted into a collection of vector using Sentence-BERT and then ingested into a standard multi layer perceptron for the classification. In this sense, *SentenceC* is a classical deep learning algorithm, as it does not contain any quantum layer during training.

SentenceQF. The second and the third pipelines are hybrid quantum-classical classifier, where the quantum layers are implemented using the *Pennylane* library [1]. Specifically, **SentenceQF** (which stands for Sentence Quantum Full) takes the vector obtained via Sentence-BERT and encodes them in a quantum circuit making use of the *Amplitude encoding* strategy: here, numerical data are encoded into the amplitude of the wavefunction (cit), i.e. in the coefficients of a linear combination of basis vectors in a complex-valued space. Hence, a set of normalized N-dimensional data points are encoded in a the coefficient of the combination of $n = log_2(N)$ qubits. This strategy allows to encode large dimensional feature vector in a relatively small number of qubit.

SentenceQS. The last pipeline implemented, named **SentenceQS** (Sentence Quantum Smooth), relies on the *Angle encoding* protocol. This strategy allows to encode a single number into an angular value of a rotation gate applied to a single qubit, for instance, given a single value, x_i^j of the jth feature vector:

$$x_i^j \rightarrow R(x_i^j)_Z|0\rangle = e^{-x_i^j \sigma_Z/2}, \tag{1}$$

where σ_Z is the Pauli Z operator. The rotations are periodic up to a global phase, hence, when using this protocol, data are conveniently normalized between $[0, \pi]$. The down side of this encoding lies on the fact that, in principle, the number of qubit needed is equivalent to the dimension of the feature vector $n = N$, thus preventing its use on currently available devices. However, it is possible to smooth this requirement by performing a dimensionality reduction on the feature vectors extracted with Sentence-BERT: this dimensionality reduction can be achieved either via standard PCA or via a fully connected neural network placed at the input of the encoding quantum circuit. Both quantum encoding strategies provide a mean to insert classical data into quantum states, which are then used for quantum computation.

Information encoded in the quantum state is processed by a parametrized quantum circuit, generally called variational ansatz, whose parameters are iteratively updated via a classical optimizer via gradient descent. Designing an optimal ansatz for quantum machine learning tasks is still an open problem: however, it is generally fundamental to find the right balance between an expressive citcuit, i.e. containing an important number of parameters and entangling gate, so that the space of solutions can be properly explored, while minimizing the circuit depth. The last requirement is rather fundamental for deploying quantum algorithms on currently available devices, as an excessive circuit depth can increase

the overall noise and hence hinder the computation. In this work, the same quantum variational ansatz is used for both *SentenceQF* and *SentenceQS*, which is a **BasicEntangledLayer** provided by Pennylane. This ansatz consist in layers of rotation gates with a trainable parameter, applied on each qubit involved, followed by a set of CNOT gates applied on pairs of consecutive qubits. Such chain of CNOT generates entanglement among the qubits involved, thus allowing to explore, while training, large sectors of the energy landscape defined by the loss function. The result of the computation is obtained performing measurements on all or a sub-set of the qubits, after the application of the parametrized circuit. The measurements outcome is thus used to evaluate a problem specific loss function, and then passed to a classical optimizer, which updates the weights both of the classical and of the quantum layers of the computation.

4 Results and Discussion

For convenience, the binary data corresponding to the accepted phrase (1) and the rejected phrase (0) are one-hot encoded into a two dimensional tensor. The embeddings from the pre-trained Sentence-BERT are properly standardized to facilitate the convergence during training. Given the strong class unbalance (approximately three times more 1 than 0), the loss function is appropriately weighted, in order to prevent bias in the classification phase. Three different experiments have been performed for the Acceptability Judgements on the Itacola Dataset. For each of them, the main properties, hyperparameters and training strategies are listed in the following:

- **SentenceC**: for the full classical protocol, which serves as a benchmark for comparing the quantum strategies, the embedding extracted from Sentence-BERT are fed into a Multi-Layer perceptron, composed of three layers, each of them with an *ReLu* activation function, with input-output dimension respectively of $768 - 384$, $384 - 192$, $192 - 96$. Last, a classification layer is applied, with a *Softmax* activation function, and dimension of the layer $96 - 2$. The whole training is performed with an Adam optimizer, with learning rate 0.01, batchsize of 16 and a weighted categorical cross entropy as objective function.
- **SentenceQF**: here the embeddings extracted from Sentence-BERT are encoded in a quantum circuit using an Amplitude encoding strategy. To encode all the 768 dimensions of the BERT feature vectors, the minimum number of qubit required is $n = \lceil log_2(768) \rceil = 10$, where the rest of the 1024 amplitudes are padded all with 0.01. After the encoding phase, the state obtained is used as an input for a two layers of parametrized *BasicEntangledLayer*. The outcome from the measurements are then passed to a Multi-Layer perceptron with a *Softmax* activation function, and input-output dimension of $10-2$, for the classification. The whole training is performed with an Adam optimizer, with learning rate 0.01, batchsize of 16 and a weighted categorical cross entropy as objective function.

– **SentenceQS**: here the embeddings extracted from Sentence-BERT are encoded first in a Multi-layer perceptron for dimensionality reduction. The network is made of three layers, each of them with an *ReLu* activation function, with input-output dimension respectively of 768−384, 384−192, 192−10. Then a quantum circuit for Angle encoding is used, with $n = 10$ qubits, that encodes the classical data into the Z-rotation gates angles. After the encoding phase, the state obtained is used as an input for a two layers of parametrized *BasicEntangledLayer*. The outcome from the measurements are then passed to a Multi-Layer perceptron with a *Softmax* activation function, and input-output dimension of $10 - 2$, for the classification. The whole training is performed with an Adam optimizer, with learning rate 0.01, batchsize of 16 and a weighted categorical cross entropy as objective function.

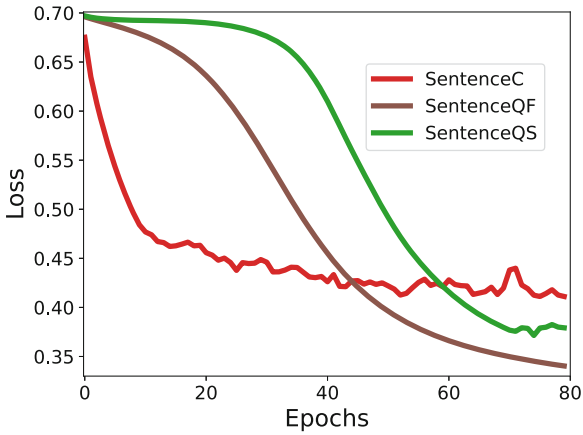


Fig. 2. Experimental results of the training for each pipeline. Training on the ItaCola Dataset for the acceptability judgements. The *SentenceC* stands for the full classical pipeline, where the embeddings from Sentence-BERT are inputs of a multi-layer perceptron designed for binary classification. *SentenceQF* refers to the quantum pipeline with amplitude embedding, where each component of the feature vector is encoded in the amplitude of a quantum state made of $n = 10$ qubit. *SentenceQS* refers to the smooth-quantum pipeline, i.e. an Angle encoding approach. A small multi layer perceptron is used before the quantum encoding to reduce the dimensionality of the feature vector from 768 to 10, so that the new data can be accommodated in the single qubits rotational gates, with $n = 10$ qubits overall.

Results of the various training are showed in Fig. 2: all the aforementioned strategies are trained for 80 Epochs. It is evident from the training phase that, while the *SentenceC* strategy reaches convergence faster compared to the quantum approaches, the steady value of the loss function is higher, indicating that the learning capacity of the model is generally smaller than the other considered. Concerning the two quantum strategies in use, the loss function in Fig. 2 indicates that, when the total information extracted from Sentence-BERT is used

in the quantum computation (as in *SentenceQF*), the training is not only faster compared to the *SentenceQS* approach, but the loss reaches a global minimum, stating an overall stronger learning ability. Conversely, when considering *SentenceQS*, the dimensionality reduction operated by the Multi-Layer percepetron and the subsequent implementation of a reduced encoding in the quantum circuit, hinder the computation, decreasing the leaning capabilities of the model. It is worth to notice that in this case the loss requires more epochs to reach convergence: this is due to the fact that, adding a whole classical layer to the quantum circuit, generates a more complex energy landscape, making the training, i.e. the finding of the optimal minimum, expensive and complicated. In this sense, it is here pointed out, that is generally advisable, when using quantum computing for transfer learning, to make use of the most information as possible, coming from the pre-trained models. Hence, it is necessary to use in this regards a proper quantum encoding strategy that prevents information loss during the computation.

Table 2. Classification results on the ItaCoLA test set using LSTM and BERT and the three training strategies: SentenceC, SentenceQF, SentenceQS.

Model	Accuracy
LSTM	0.794
ITA-BERT	0.904
SentenceC	0.67
SentenceQF	0.81
SentenceQS	0.71

The performances of each training are captured by the accuracy of the classification on the test set. The accuracy might be biased by the presence of class unbalance: however, both the weighting in the loss function and the balanced sampling on the test set are able to prevent biased classification. The results are listed in Table 2: while *ITA-BERT* is still the most accurate model for the Acceptability Judgement task, both the quantum classifier in use, with transfer learning from Sentence-BERT, show improved performances compared to the classical pipeline *SentenceC*. Furthermore, *SentenceQF* is able to outperform quite significantly the LSTM. This is a promising result for two particular reasons: first it shows the potential of quantum computing in general terms for machine learning tasks. In fact, compared to an LSTM, the *SentenceQF* model has fewer parameters but it still able to learn more efficiently. Second it shows that hybrid quantum-classical methods can play already an important role for NLP tasks. The ability of quantum circuit to represent language data in complex high dimensional vector spaces can fuel the most sophisticated NLP deep learning model with higher expressive capabilities, hence potentially impacting the performances of task beyond the binary classification hereby investigated.

5 Conclusion

In this work, a novel approach for transfer learning with quantum computing has been investigated, focused on the Acceptability Judgment task on Italian language using an annotated dataset. Specifically, using a whole sentence embedding strategy, provided by Sentence-BERT, a quantum classifier has been trained, with two different encoding strategies. Results demonstrate the potential of this approach: while the metrics are not yet comparable with the most advanced classical models, such as ITA-BERT, they have the advantage of being good enough even with a relatively simple ansatz. Furthermore, they show an advantage on the specific task considered both over a simple multilayer perceptron, and when compared with more complicated benchmarks, such as the LSTM.

Future works on the same pipelines hereby proposed, will focus on two elements: first, overcoming the sentence based embedding, provided by sentence-BERT, and focusing on single word embedding that in principle entail more information about the semantics and the properties of each element of the phrase. Second, on the quantum level, increasing the complexity of the ansatz in order to rise up the overall expressivity of the model. These two elements combined might be able to show quantum advantage on NLP transfer learning tasks. From a strictly linguistic point of view, the prediction and classification of acceptability judgments in NLP play a crucial role in improving the quality and fluency of various NLP applications, ranging from language models to language generation. Overall, By advancing the understanding and modeling of acceptability judgments, researchers aim to enhance the effectiveness and naturalness of human-computer interactions, making NLP systems more linguistically informed and aligned with human expectations.

Although still in its infancy and with numerous hardware limitations, quantum computing in NLP classification tasks offers a novel perspective on addressing NLP challenge. Quantum-inspired algorithms, quantum embeddings, and quantum machine learning models present exciting avenues for improving classification accuracy, semantic understanding, and computational efficiency. Further research and development in this interdisciplinary field have the potential to unlock new possibilities for quantum-assisted natural language processing.

Acknowledgments. We acknowledge financial support from the project PNR MUR project PE0000013-FAIR.

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